

1 Machine Learning Introduction

1.1 Telugu Transliteration lo Explain Like I'm 10 Guide

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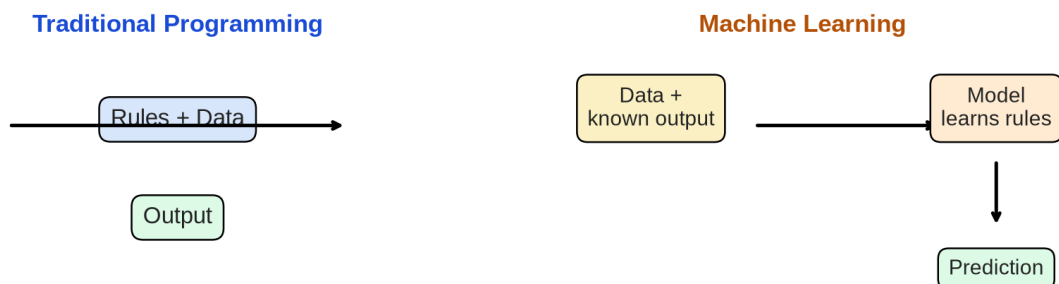
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One-line idea: Machine Learning ante computer ki examples chupinchi, data lo patterns nerpinchi, kotha examples meeda prediction or decision cheyinchadam.

1.2 1. Machine Learning ante emiti?

Machine Learning (ML) anedi Artificial Intelligence lo oka important branch. Traditional programming lo manam rules manually write chestam. Machine Learning lo manam data and known answers istam; computer aa data nundi useful patterns ni automatic ga learn chestundi.

Two ways computers solve problems



1.2.1 Traditional programming

Rules + Data → Output

Example: temperature 35°C kanna ekkuva and humidity 80% kanna ekkuva aithe “Very uncomfortable” ani print cheyyadam. Ee rule manishi manually rayali. Rules miss aithe output kuda miss avutundi.

1.2.2 Machine Learning

Data + Known Output → Model learns rules → New prediction

Example: thousands of past records lo temperature, humidity, and comfort rating unte model relationship ni learn chestundi. Future lo kotha temperature and humidity ichinappudu comfort rating predict chestundi.

1.2.3 Baby learning analogy

Baby ki cats pictures and non-cat pictures chala chupiste, baby cat ki common patterns observe chestundi. Kotha picture chusi cat aa kaada ani cheppadaniki try chestundi. ML model kuda training examples nundi pattern learn chesi unseen data meeda prediction chestundi.

1.2.4 Student exam analogy

Student past questions and solutions practice chestadu. New exam question vachinappudu memorized question kaakunda, nerchukunna pattern use chesi solve chestadu. ML model lo training data past questions laantidi; test data new exam laantidi.

Important: ML simply “data ni memorize cheyyadam” kaadu. Goal is new, unseen data meeda generalize cheyyadam.

1.3 2. Machine Learning ela work chestundi?

1.3.1 Step 1: Data collect cheyyadam

Past sales, patient records, images, sensor readings, emails, or customer events collect chestam. Data relevant and representative ga undali.

1.3.2 Step 2: Data prepare cheyyadam

Missing values, duplicate rows, outliers, wrong formats, and inconsistent units handle chestam. Features select or engineer chestam. Some models ki scaling kuda avasaram.

1.3.3 Step 3: Algorithm choose cheyyadam

Problem type, data size, feature format, interpretability, and deployment constraints batti algorithm choose chestam. Examples: Linear Regression, Logistic Regression, Decision Tree, SVM, Random Forest, Neural Network.

1.3.4 Step 4: Model train cheyyadam

Algorithm training data ni chusi internal parameters learn chestundi. Ee parameters examples nundi automatic ga adjust avutayi.

1.3.5 Step 5: Model evaluate cheyyadam

Unseen validation or test data meeda predictions chesi suitable metric calculate chestam. Regression ki MAE/RMSE; classification ki precision, recall, F1, ROC-AUC laanti metrics use cheyyachu.

1.3.6 Step 6: Tune and improve cheyyadam

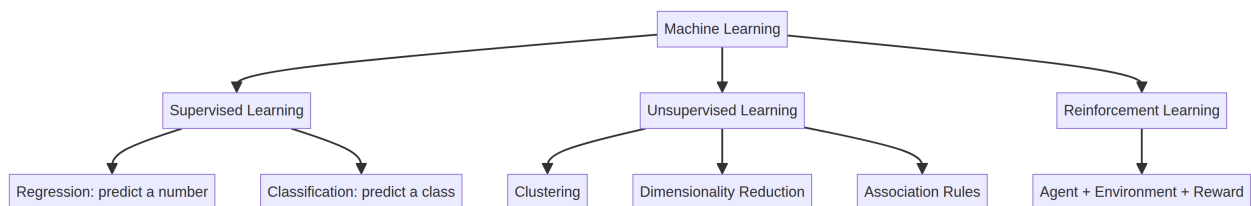
Hyperparameters, features, data quality, model choice, and threshold improve chestam. Test set ni repeated tuning kosam use cheyyakandi.

1.3.7 Step 7: Deploy cheyyadam

Model ni application, API, dashboard, batch job, or device lo use chestam. Deployment tarvata latency, drift, data quality, and performance monitor cheyyali.

1.3.8 Memory workflow

Collect → Prepare → Choose → Train → Evaluate → Improve → Deploy → Monitor



1.4 3. Enduku Machine Learning use cheyyali?

Manual rules chala simple problems ki useful. Kani konni problems lo rules manually rayadam impractical.

- **Image recognition:** Millions of pixels and complex shapes.
- **Natural Language Processing:** Words context batti meaning marutundi.
- **Fraud detection:** Fraud patterns rare and continuously change avutayi.
- **Medical diagnosis:** Many symptoms and test values combination.
- **Recommendation systems:** User behavior and item relationships.
- **Genomics:** Very large biological datasets.
- **Satellite imagery:** Huge images and spatial patterns.
- **Spam detection:** Spammers rules ni evade cheyadaniki language marchutaru.

ML data ekkuva unte automatically useful patterns find cheyyagaladu. Kani “more data” alone sufficient kaadu. Data quality, labels, fairness, leakage, and monitoring kuda important.

1.5 4. Machine Learning main types

1.5.1 4.1 Supervised Learning

Supervised Learning lo each training example ki input features and correct output label rendu untayi. Teacher answer key istunnattu.

- House features + known price → price pattern.
- Email words + spam/not spam → spam classifier.
- Symptoms + disease label → diagnosis-support model.

Supervised Learning two major tasks:

1. **Regression:** continuous number predict cheyyadam.
2. **Classification:** category or class predict cheyyadam.

1.5.2 4.2 Unsupervised Learning

Unsupervised Learning lo labels or correct answers levu. Model data lo natural structure ni discover chestundi.

- Similar customers ni groups ga divide cheyyadam.
- Many features ni fewer dimensions lo represent cheyyadam.
- Transactions lo items together occur avvadam discover cheyyadam.

Main forms: clustering, dimensionality reduction, and association rules.

1.5.3 4.3 Reinforcement Learning

Reinforcement Learning lo agent environment tho interact avutundi. Good action ki reward, bad action ki penalty or low reward. Trial and error dwara long-term reward maximize cheyyadam goal.

1.5.4 4.4 Other learning settings

- **Semi-supervised learning:** small labeled dataset + large unlabeled dataset.
- **Self-supervised learning:** data nundi model itself training target create chestundi. BERT masked-word prediction example.

1.5.5 Types ni compare cheyyandi

Type	Labels?	Main question	Examples
Supervised	Yes	“Correct answer ni predict cheyyagalama?”	Price, spam, disease class
Unsupervised	No	“Data lo natural structure emiti?”	Customer groups, PCA
Reinforcement	Reward signal	“Next best action emiti?”	Games, robotics
Semi-supervised	Few labels	“Limited labels ni expand cheyyagalama?”	Medical images
Self-supervised	Self-created target	“Input lo missing/next part predict cheyyagalama?”	Masked words, next token

1.6 5. Supervised Learning

1.6.1 Key idea

Input Features + Output Label → Model learns mapping

Training lo known input-output examples use chestam. Prediction time lo kotha input isthe model output estimate chestundi.

1.6.2 Examples

Input	Output	Problem
House size, location	Price	Regression
Email words	Spam / Not spam	Classification
Patient symptoms	Disease class	Classification
Past sales	Future sales	Regression
Image pixels	Cat / Dog	Classification
Study hours	Exam score	Regression

1.7 6. Regression: “How much?” or “How many?”

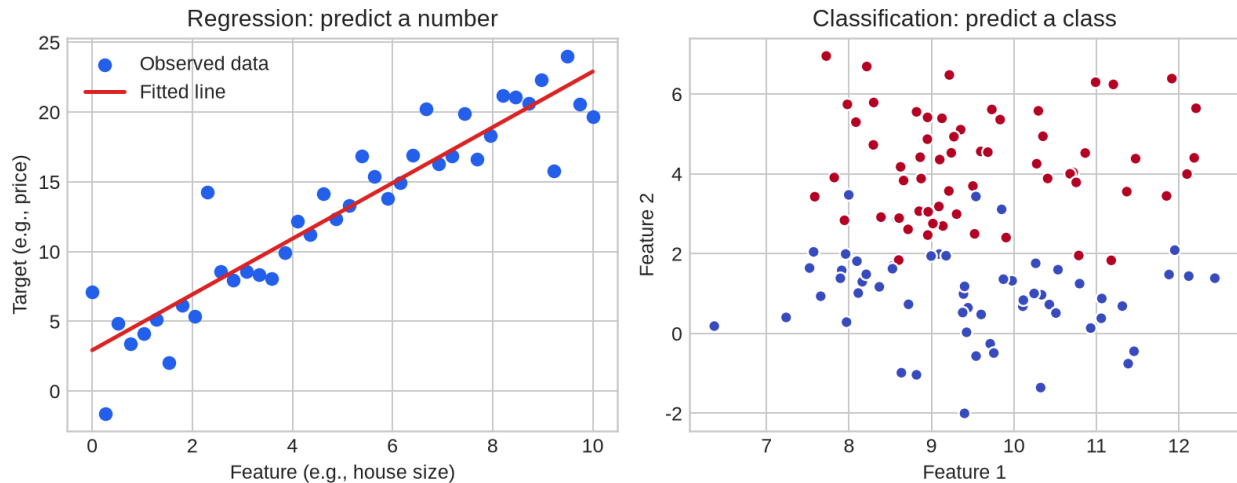
Regression continuous numerical value predict chestundi. Output any reasonable number avvachu.

Examples:

- House price: ₹45 lakhs.

- Tomorrow temperature: 32°C.
- Next month sales: 1,247 units.
- Delivery time: 38.5 minutes.

Supervised learning outputs



1.7.1 Regression algorithms

Algorithm		Eppudu useful?		Non-linear relation?	Main note
Simple Regression	Linear	One feature,	roughly straight relation	No	Easy baseline
Multiple Regression	Linear	Many features	numeric	No	Coefficients interpret cheyyachu
Polynomial Regression		Curved relation		Yes	High degree overfit avvachu
Ridge		Correlated features,	regularization	Usually linear	L2 penalty
Lasso		Irrelevant features	ekkuva	Usually linear	L1 feature selection cheyyagaladu
Elastic Net		Ridge + Lasso	behavior	Usually linear	Mixed penalty
SVR		Flexible margin-based	regression	Yes with kernel	Scaling important
Decision Tree		Rule-like non-linear	pattern	Yes	Depth control important
Random Forest		Robust tabular	baseline	Yes	Many trees ensemble

Algorithm	Eppudu useful?		Non-linear relation?	Main note	
Gradient Boosting	Strong performance	tabular	Yes	Sequential correction	error
Neural Network	Very relationships	complex	Yes	More data and tuning often needed	

1.7.2 Regression metrics

Let actual value y_i , prediction $\hat{y}_i$, and number of rows n .

1.7.2.1 MAE: Mean Absolute Error

$$[MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|]$$

Actual values [100, 200], predictions [90, 230]. Errors are 10 and 30. $MAE = (10+30)/2 = 20$. Ante average ga 20 units off.

MAE easy to interpret and same unit lo untundi. Very large errors ki MSE kanna less sensitive.

1.7.2.2 MSE: Mean Squared Error

$$[MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2]$$

Same example: $MSE = (10^2 + 30^2)/2 = (100+900)/2 = 500$. Large error square avvadam valla outliers ki strong penalty.

1.7.2.3 RMSE

$$[RMSE = \sqrt{MSE}]$$

MSE 500 aithe $RMSE \approx 22.36$. Output same unit lo untundi. Large mistakes ni emphasize chestundi.

1.7.2.4 R-squared

$$[R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}]$$

$R^2=1$ perfect prediction. $R^2=0$ target mean ni predict cheyyadam kanna better kaadu. $R^2<0$ mean baseline kanna worse. R^2 “85% accuracy” ani kaadu; target variance lo model explain chese fraction.

1.7.2.5 MAPE

$$[MAPE = \frac{100}{n} \sum \left| \frac{y_i - \hat{y}_i}{y_i} \right|]$$

Percentage interpretation easy. Kani actual value zero or near zero aithe MAPE problematic.

1.7.3 Regression code example

```
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.linear_model import Ridge
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

X, y = load_diabetes(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```

model = Ridge(alpha=1.0)
model.fit(X_train, y_train)
predictions = model.predict(X_test)

mae = mean_absolute_error(y_test, predictions)
rmse = np.sqrt(mean_squared_error(y_test, predictions))
r2 = r2_score(y_test, predictions)
print("MAE:", round(mae, 2))
print("RMSE:", round(rmse, 2))
print("R-squared:", round(r2, 3))

```

alpha Ridge regularization strength. Alpha chala high aithe model coefficients too much shrink avvachu; chala low aithe regularization effect weak ga untundi. Dataset-specific tuning cheyyali.

1.8 7. Classification: “Which class?”

Classification fixed categories predict chestundi.

- Email → spam or not spam.
- Tumor → malignant or benign.
- Image → cat, dog, or bird.
- Digit → 0 to 9.

1.8.1 Classification types

1.8.1.1 Binary classification

Exactly two classes: fraud/legitimate, pass/fail, disease/no disease.

1.8.1.2 Multi-class classification

Three or more mutually exclusive classes: cat/dog/bird, Iris flower classes, digits 0–9.

1.8.1.3 Multi-label classification

One sample ki multiple labels possible: movie = action + comedy; article = politics + economy; image = sky + mountain + snow.

1.8.2 Classification algorithms

- **Logistic Regression:** probability-based linear boundary; strong baseline.
- **Decision Tree:** if-then rules; easy to explain.
- **SVM:** maximum-margin boundary; scaled data useful.
- **Random Forest:** many trees voting; robust tabular model.
- **Gradient Boosting:** sequential correction; often strong, but not always best.
- **KNN:** nearest examples vote; scaling important.
- **Naive Bayes:** probability-based and fast, especially text baselines.
- **Neural Network:** complex patterns, images, text, and large data.

1.8.3 Classification code example

```
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix

X, y = load_breast_cancer(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, stratify=y, random_state=42
)

model = Pipeline([
    ("scale", StandardScaler()),
    ("classifier", LogisticRegression(max_iter=2000))
])
model.fit(X_train, y_train)
predictions = model.predict(X_test)

print(confusion_matrix(y_test, predictions))
print(classification_report(y_test, predictions))
```

1.8.4 Important metrics

- **Accuracy:** all correct / all predictions. Balanced classes unnapudu useful.
- **Precision:** predicted positive lo nijamaina positive proportion.
- **Recall:** actual positive lo model catch chesina proportion.
- **F1-score:** precision and recall balance.
- **ROC-AUC:** ranking discrimination.
- **PR-AUC:** positive class rare unnapudu often useful.
- **Confusion matrix:** TP, TN, FP, FN counts.

Medical screening or fraud detection lo accuracy high unna recall poor aithe dangerous avvachu. Metric business cost batti choose cheyyali.

1.9 8. Unsupervised Learning

Unsupervised Learning lo correct labels levu. Model data lo natural patterns find chestundi.

1.9.1 Warehouse sorting analogy

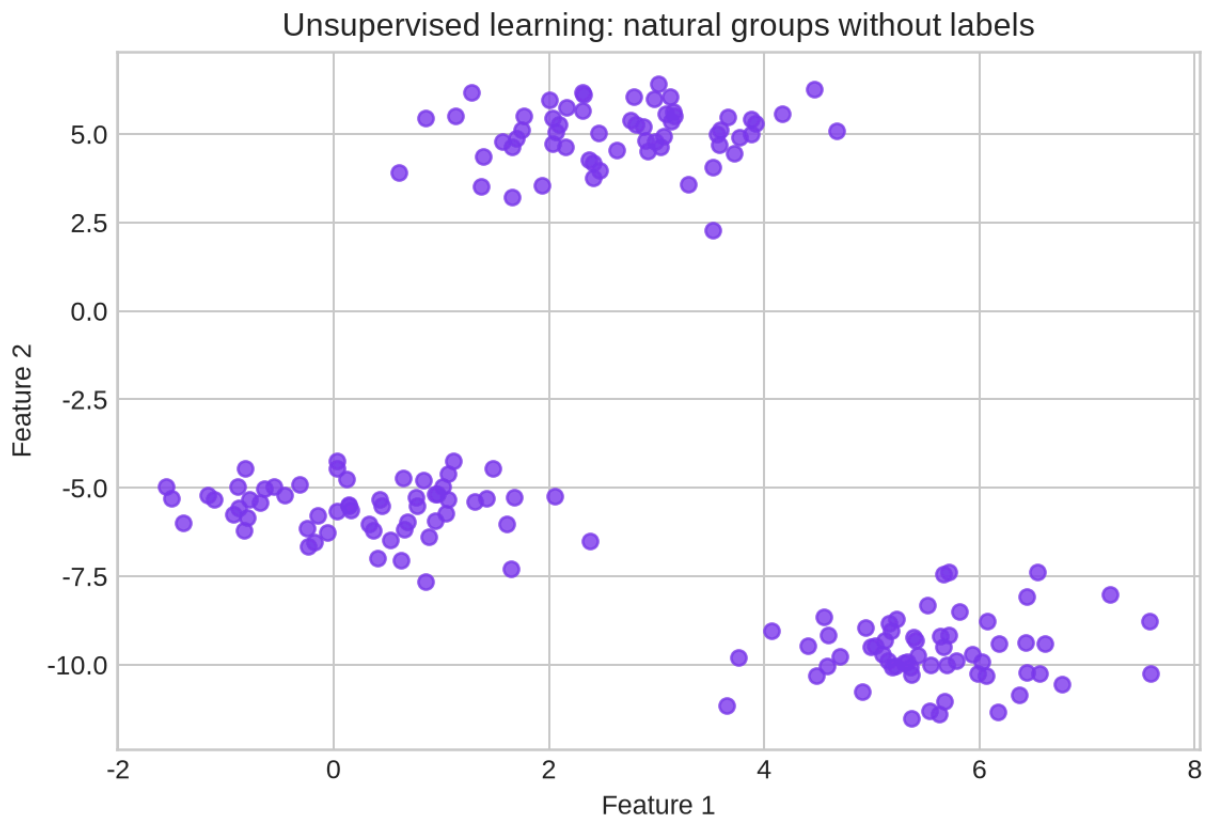
Random boxes tho warehouse worker ki labels or instructions levu. Worker size, weight, and color chusi similar boxes groups ga arrange chestadu. Unsupervised model kuda input patterns chusi structure discover chestundi.

1.9.2 8.1 Clustering

Clustering similar data points ni groups ga form chestundi. Groups munduga labels ga ivvamu.

Uses:

- Customer segmentation.
- Product grouping.
- Gene expression analysis.
- Anomaly discovery.



1.9.2.1 K-Means idea

K-Means lo K cluster centers (centroids) choose chesi, points ni nearest centroid ki assign chestam. Centroids repeatedly update chestam. Goal within-cluster distances reduce cheyyadam.

Important caution: K-Means spherical-like clusters and numeric scaled features ki better. K choose cheyyali. Outliers and different density shapes lo DBSCAN consider cheyyachu.

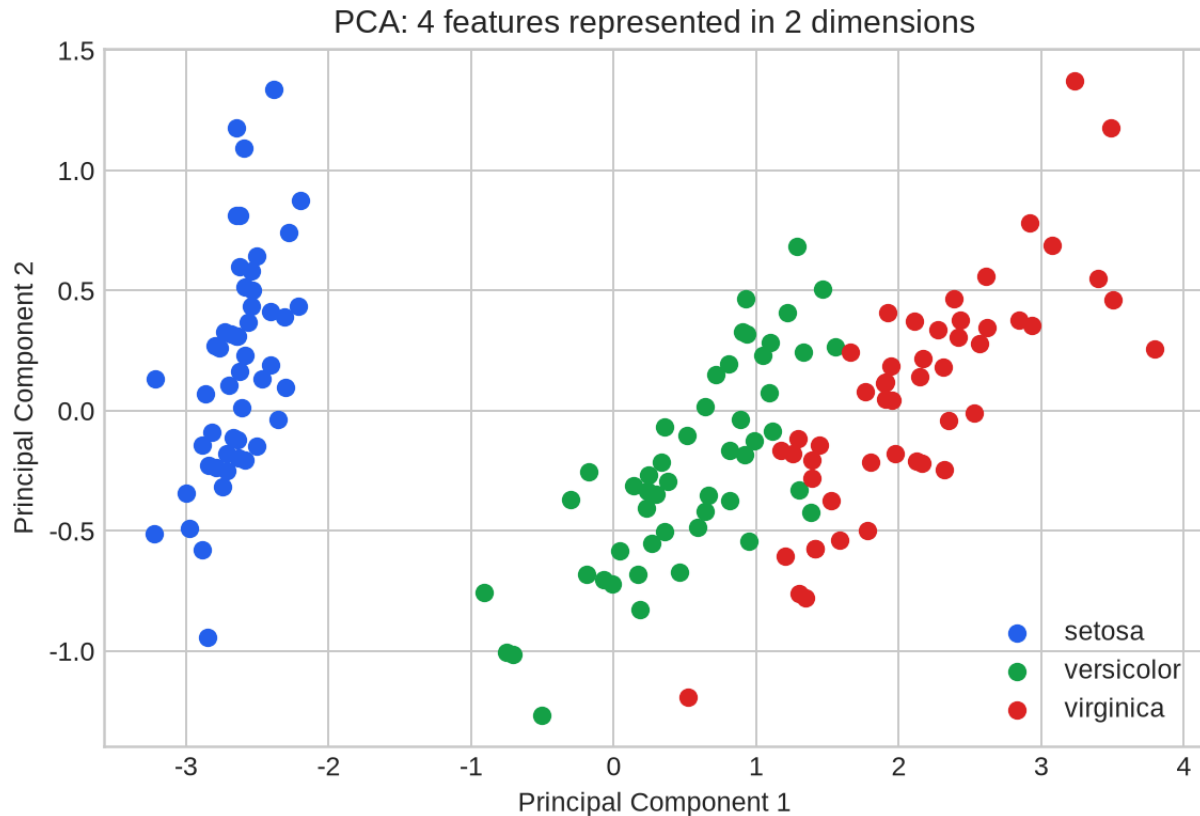
1.9.2.2 DBSCAN

Density-based algorithm. Arbitrary shape clusters find cheyyagaladu and noise points identify cheyyagaladu. eps and min_samples sensitive settings.

1.9.3 8.2 Dimensionality Reduction

Many features ni fewer dimensions lo represent cheyyadam. Goal important information maximum possible retain cheyyadam.

Map analogy: Earth 3D information ni 2D map lo compress chestam. Height information konchem lose avvachu, kani map useful ga untundi.



1.9.3.1 PCA idea

PCA data variance ekkuvaga unde directions ni principal components ga choose chestundi. First component maximum variance capture cheyyadaniki try chestundi; second component first ki perpendicular ga next most variance capture chestundi.

Uses:

- High-dimensional visualization.
- Noise reduction.
- Faster modeling.
- Correlated features compress cheyyadam.

PCA components interpretability original features laga direct ga undakapovachu. Scaling before PCA often important.

1.9.4 8.3 Association Rule Learning

Transactions lo items together occur avvadam discover chestundi. Supermarket lo “bread konnavaallu butter kuda konnaru” pattern identify chesi butter ni bread daggara place cheyyachu.

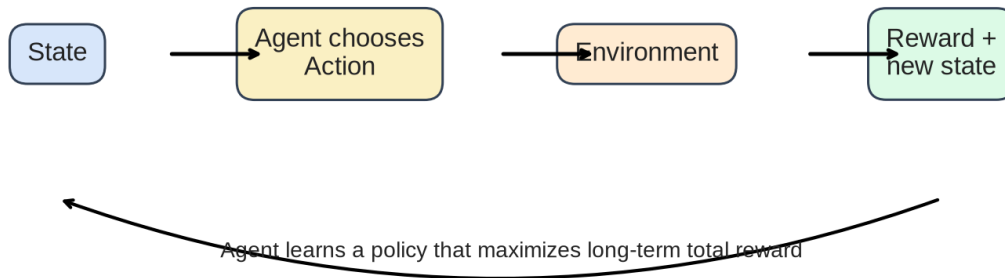
Important concepts:

- **Support:** Rule or itemset entha frequently occur avutundi.
- **Confidence:** X unna transactions lo Y kuda entha often untundi.
- **Lift:** X and Y together occurrence independent chance kanna entha ekkuva?

Association means co-occurrence; causation ani automatically conclude cheyyakandi.

1.10 9. Reinforcement Learning

Reinforcement Learning (RL) lo agent environment tho interact avutundi and actions try chestundi.



1.10.1 Components

- **Agent:** decision maker; robot or program.
- **Environment:** agent interact ayye world.
- **State:** current situation.
- **Action:** agent cheyyagalige move.
- **Reward:** action quality feedback.
- **Policy:** state batti action choose chese strategy.

1.10.2 Cycle

State → Action → Environment response → Reward + new state → repeat

Goal immediate reward matrame kaadu. Time over total reward maximize cheyyadam. Games, robotics, traffic control, and resource allocation lo RL use cheyyachu.

1.10.3 Dog-training analogy

Dog sit aithe treat reward. Sit cheyyakapothe reward ledu. Repeated practice lo dog “sit → treat” relation nerchukuntundi. RL lo reward design chala important; wrong reward function aithe agent unintended behavior learn cheyyachu.

1.10.4 RL caution

Reward signal labels laga direct correct answer kaadu. Exploration vs exploitation, safety, sample cost, and delayed reward problems consider cheyyali.

1.11 10. Semi-supervised and self-supervised learning

1.11.1 Semi-supervised learning

Small amount labeled data and large amount unlabeled data combine chestundi. Human labeling expensive; raw unlabeled data collect cheyyadam cheap.

Typical process:

1. Small labeled dataset meeda initial model train.
2. Unlabeled examples ki pseudo-labels generate.

3. High-confidence examples add chesi retrain.
4. Validation and quality checks continue cheyyandi.

Uses: medical imaging, text classification, web content classification.

Pseudo-labels wrong aithe error reinforce avvachu. Confidence threshold and human review useful.

1.11.2 Self-supervised learning

Model unlabeled data nundi own training task create chestundi.

- BERT: masked word ni context nundi predict cheyyadam.
- Video: previous frames nundi next frame predict cheyyadam.
- LLM: previous tokens nundi next token predict cheyyadam.

Self-supervised pretraining tarvata supervised fine-tuning or task-specific evaluation cheyyachu.

1.12 11. Which type eppudu use cheyyali?

Question	Suitable approach
“How much?” / “How many?”	Regression
“Which class?”	Classification
“Natural groups emiti?”	Clustering
“Data ni simple ga represent cheyyacha?”	Dimensionality Reduction
“Which items appear together?”	Association Rules
“Next action emiti?”	Reinforcement Learning
Few labels, many raw examples?	Semi-supervised / self-supervised

1.12.1 Model selection decision tree

1. Output continuous number aithe regression.
2. Output fixed class aithe classification.
3. Labels levu, groups kavali aithe clustering.
4. Features chala ekkuva, visualization or compression kavali aithe PCA or another dimensionality-reduction method.
5. Transaction co-occurrence kavali aithe association rules.
6. Repeated action and feedback unte reinforcement learning.

1.13 12. Common mistakes

Mistake	Correct approach
Test data training lo mix cheyyadam	Train/validation/test separation maintain cheyyandi
Preprocessing split mundu fit cheyyadam	Pipeline use cheyyandi

Mistake	Correct approach
Classification lo accuracy matrame chudatam	Precision, recall, F1, PR-AUC, cost inspect cheyyandi
Regression lo MAPE zero values tho use cheyyadam	MAPE limitations check; MAE/RMSE consider
Correlation ni causation ani cheppadam	Controlled analysis or domain evidence kavali
Clustering groups “true” ani assume cheyyadam	Scaling, algorithm assumptions, stability validate cheyyandi
PCA ni feature selection ani confuse cheyyadam	PCA new components create chestundi; original features select cheyyadu
RL lo reward wrong design	Safety constraints and unintended behavior tests add cheyyandi
Labeled data quality ignore cheyyadam	Label consistency and bias audit cheyyandi
Model deployment tarvata monitor cheyyakapovadam	Drift, latency, missing inputs, metric monitoring setup cheyyandi

1.14 13. Practical applications

1.14.1 Finance and credit risk

Inputs: income, repayment history, debt ratio. Output: default probability or risk score. Metrics: ROC-AUC, PR-AUC, calibration, expected business cost. Explainability and fairness important.

1.14.2 Fraud detection

Inputs: transaction amount, device, location, velocity. Output: fraud probability. Recall and PR-AUC useful; false positives customer experience impact chestayi.

1.14.3 Healthcare

Inputs: symptoms, labs, imaging. Output: risk or diagnosis support. Sensitivity, calibration, external validation, and clinical review mandatory.

1.14.4 Customer churn

Inputs: usage, complaints, tenure. Output: churn probability. Model output actionability and intervention cost tho evaluate cheyyali.

1.14.5 Recommendation and search

Inputs: user behavior, item features, query. Output: ranked items. Ranking metrics, freshness, latency, and online A/B testing important.

1.14.6 Manufacturing

Inputs: sensor signals, temperature, vibration. Output: failure risk or remaining useful life. False alarms and missed failures costs balance cheyyali.

1.14.7 NLP, chatbots, and LLM systems

Inputs: text, conversation, documents. Output: class, embedding, answer, or retrieved evidence. Evaluate relevance, factuality, safety, latency, and cost.

1.15 14. Beginner-friendly Python workflow

```
# 1. Understand the task and target
# 2. Split data before fitting preprocessing
# 3. Build a Pipeline
# 4. Train only on training data
# 5. Evaluate with a metric that matches the risk
# 6. Inspect errors, not only one score

from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, stratify=y, random_state=42
)

pipeline = Pipeline([
    ("scaler", StandardScaler()),
    ("model", LogisticRegression(max_iter=2000))
])
pipeline.fit(X_train, y_train)
predictions = pipeline.predict(X_test)
```

1.15.1 Line-by-line meaning

- `train_test_split`: final unseen test portion separate chestundi.
 - `stratify=y`: class proportions preserve chestundi.
 - `Pipeline`: scaler training fold meeda matrame fit avvadam ensure chestundi.
 - `random_state=42`: repeatable split.
 - `fit`: model learns parameters.
 - `predict`: new inputs ki classes return chestundi.
-

1.16 15. Visuals ni ela read cheyyali?

1. **Traditional vs ML diagram**: Traditional lo rules manam supply chestam; ML lo examples nundi model pattern learn chestundi.

2. **Regression plot:** Points and fitted line madhya vertical gap prediction error.
3. **Classification plot:** Similar class colors natural regions or boundaries suggest chestayi.
4. **Clustering plot:** Labels ivvakapoyina nearby points groups laga kanipistayi.
5. **PCA plot:** Original 4 dimensions ni two principal components lo view chestam; visual separation useful, but information complete ga preserve ani guarantee kaadu.
6. **RL loop:** Agent state chusi action choose chestundi; environment reward and next state istundi.
7. **Taxonomy diagram:** ML ni supervised, unsupervised, and reinforcement branches ga remember cheyyandi.

Visual memory trick: Teacher answer isthe supervised; teacher lekapothe unsupervised; game lo points/penalties unte reinforcement.

1.17 16. Final revision summary

- ML = data nundi patterns learn chesi new data meeda predictions/decisions.
- Traditional programming = Rules + Data → Output.
- ML = Data + Output → Model learns rules.
- Supervised learning labels tho train avutundi.
- Regression continuous number predict chestundi.
- Classification category predict chestundi.
- Unsupervised learning labels lekunda structure discover chestundi.
- Clustering natural groups create chestundi.
- PCA dimensions reduce chestundi.
- Association rules co-occurring items discover chestayi.
- Reinforcement Learning reward and penalty dwara action policy nerchukuntundi.
- Semi-supervised = few labels + many unlabeled examples.
- Self-supervised = data itself nundi training targets.
- Evaluation metric problem risk batti choose cheyyali.
- Test data final honest evaluation kosam reserve cheyyali.
- Good ML system model matrame kaadu; data, evaluation, deployment, and monitoring kuda.

1.17.1 Night-time recall answers

- **Regression?** Number predict cheyyadam.
- **Classification?** Category predict cheyyadam.
- **Clustering?** Labels lekunda similar items group cheyyadam.
- **PCA?** Important information retain chestu dimensions reduce cheyyadam.
- **RL?** Reward feedback tho decisions nerchukovadam.

1.17.2 Interview-ready answer

“Machine Learning is a branch of AI in which a model learns patterns from data to make predictions or decisions on new examples. Supervised learning uses labels for regression and classification; unsupervised learning discovers structure through clustering, dimensionality reduction, or association rules; reinforcement learning learns actions using rewards from an environment. A reliable ML

workflow separates data, prevents leakage, selects appropriate metrics, evaluates honestly, and monitors deployment.”

1.18 References

Final takeaway: ML ni remember cheyyadaniki formula: **Good data + correct problem type + suitable model + honest evaluation + responsible deployment.**